

Answer: B

8 Max GPE = Total Energy = Max KE

$$\begin{aligned} KE_{\text{max initial}} &= \frac{1}{2} m \omega^2 x_0^2 \\ &= \frac{1}{2} (0.2) \left(\frac{2\pi}{0.5} \right)^2 (0.015)^2 \\ &= 3.6 \times 10^{-3} \text{ J} \end{aligned}$$

$$\begin{aligned} KE_{\text{max final}} &= \frac{1}{2} m \omega^2 x_0^2 \\ &= \frac{1}{2} (0.2) \left(\frac{2\pi}{0.5} \right)^2 (0.0065)^2 \\ &= 0.7 \times 10^{-3} \text{ J} \end{aligned}$$

$$KE_{\text{max initial}} - KE_{\text{max final}} = 2.9 \times 10^{-3} \text{ J}$$

Answer: B

